

Zen and the Art of the Final Year Project

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Chapter 1

Introduction

Important IP Notice

Project ownership and copyright belongs to the Institute. You should put no indication of copyright, assumed or otherwise, on any of your project documentation, except where you are including material that is the copyright of others. Note that, in the event of you wishing to further develop the project for commercial or personal use, the Institute may consider granting you a licence to do so.

All student projects must be accompanied by a written declaration that the project is not plagiarised and has been completed in accordance with college regulations. A standard declaration form is available.

What is the Final Year Project (FYP)

The FYP is a capstone project. It serves to bring together all the different modules that are taught on the course. Completing the FYP will require that you not only master the individual subjects but also to be able to integrate all these subjects into a coherent whole. It is considered the most important part of the course and as a result carries significantly more weight when calculating your final graduation grade. This can be seen in the amount of credits awarded to the project (FYP carries twenty credits while all other subjects carry either ten or five credits - i.e. half or one quarter of the FYP).

Successful completion of the FYP demonstrates that the student has:

- achieved mastery of all the individual course modules;
- integrated this knowledge into a cohesive and integrated whole;

- Applied this knowledge on a large scale software development project.

Employers look to FYP when deciding whether to offer a position to a graduate because it is here that graduates demonstrate their ability to practically apply their skills in a work setting.

Successful completion of the FYP requires the following skills:

Integration of Knowledge The courses you do are taught as individual stand alone subjects. The project is where you show that you understand how all these topics are related. FYP is where you bring them together and demonstrate your holistic understanding of all areas of software development.

Practical Application of Theory The project allows you to take the theory you have learnt and demonstrate that you can apply it in practice.

Ability to work in novel areas Software development is a rapidly changing field. During your career you will have to learn new techniques, technologies and processes. Each project has at least one novel aspect. FYP will teach you how to research new areas on your own and incorporate this into your existing knowledge.

Management of Development Process Managing a large development project is difficult. The only way to learn how to do this is to actually manage a large development project such as the FYP.

Familiarity with Development Toolset You will need to become familiar and comfortable with the range of tools used in modern software development.

Chapter 2

Role of Project Supervisors/Mentors

Every project has an assigned supervisor. The supervisor will be a lecturer who has experience with the software development process. Supervisors set the initial project specification and so will have a certain amount of expertise and an interest in the project topic area. Occasionally students set their own projects or undertake projects set by external companies¹. In this case the lecturers who supervise these projects is decided in consultation with the companies and/or the students. In any case all supervisors are committed to helping the students complete a successful FYP and have the knowledge required to advise you.

The supervisor plays an advisory role in the FYP. A more correct title for project *supervisors* might be project *mentors*. The project does not belong to the mentor/supervisor but to the student. You must take ownership of the project for yourself. Your mentor will act as advisor and sometimes (if necessary) take the role of the project client to help guide the student.

Your mentor will advise on all aspects of the project development and will try ensure that the project meets the expected standards. She will also help ensure that the correct project deliverables are produced within the appropriate timeframe. Remember you are ultimately responsible for your own project not your mentor. This means that all decisions made during the development process must be made by you, the mentor only advises and you can choose to follow or ignore that advice. Of course, given the familiarity of the mentors with the FYP process you must be very sure of yourself before ignoring any advice that you are given.

¹Usually companies where students have spent the summer working on work placement.

Chapter 3

Acceptable Project Proposals

FYP Specification Format

The initial FYP proposal contains the following information:

Title FYP Title (one line)

Brief Description A short one paragraph overview of the project idea.

Technologies A list of the technologies that will need to be mastered in order to implement the FYP.

Full Description A more detailed description of the project functionality and purpose. It will contain enough technical details for the student to research the project idea and decide if they are interested in it.

Mandatory A list of FYP features (or use cases) that must be implemented in order to get a pass grade.

Discretionary A list of features (or use cases) that must be completed in order to get an honours grade.

Exceptional A list of features (or use cases) that must be completed in order to get a first class honours grade.

Choosing a FYP

FYP specifications are made available to all students at the start of the year. Students will have one week to examine the proposed specifications and decide on their choice. During this week students are expected to research the projects and approach and talk to the mentors associated with any FYP they are considering. At the end of the week each student emails a list of the

projects that they want to undertake. This list of projects must be in order of preference. Once the full set of FYP preferences are received the projects will be allocated to students in as fair a manner as possible. The general guideline applied are:

1. Starting with the first preferences every student who picks a FYP as their number one preference that no one else has picked will get that FYP.
2. If two or more students choose the same FYP at the same preference level then one will be assigned that FYP and the others will move to their next preference.
3. After all first preference projects are assigned the second preference FYP's will be assigned in the same manner.
4. This continues until everyone is assigned a project.
5. When deciding who gets assigned a FYP that multiple students have voted for at the same level of preference an effort is made to make sure that the choice will result in the most possible students getting a FYP as close to their number one choice as possible. For example, if Mary and Bert both have the same project picked as their first preference then if when Mary is given this project it results in Bert getting his second choice but if when Bert is given his first preference Mary is left with her third choice, then in this case Mary will be given her first preference.
6. It is in the students interest to submit as many preferences as possible to avoid being assigned a FYP at random.
7. Once projects have been assigned they will not be changed.

While it is important to get a project that you want it is inevitably the case that some students will end up getting a project that is not of their choice. The procedure above is in place to ensure that the number of students not getting their choice is as low as possible. Once you are assigned a project, even if it was not your first choice, you need to take responsibility for that project. You must be the project driver not the supervisor.

Sourcing FYP Proposals

FYP proposals come from three sources:

Mentor Specified Projects These are projects that mentors are interested in running. Most FYP proposals come from this source.

Company Specified Projects Most of these ideas come from companies that have third year students on work placement. In most cases they want the student to continue working with them during their final year of the degree and often plan on employing those students after they graduate. Ocassionally companies with no previous association with the college will offer to specify projects of interest to them in order to assess the quality of our course. In these cases a sucessful FYP tends to lead to a job offer for the student. In either case the FYP specification is checked by the project mentors to ensure it is at an appropriate standard for this course and any IP issues are resolved in advance.

Student Specified Projects Some students have ideas of their own that they wish to develop. These projects can usually be turned into acceptable specifications by the mentors who will make sure that they are of the appropriate standard. In some cases a student proposed project may not be accepted by any supervisor. This is due to either supervisors not feeling that they know enough about the area to mentor this FYP or else being unable to mould the specification into an acceptable FYP. If you are unsure of your idea it is best to approach the supervisors and ask for advice on your idea.

Example FYP Specification

Wearable using Movesense hardware

Brief Description

Wearable that employs the Movesense wearable sensor.

Technologies

1. Android and/or PC
2. C++17
3. Movesense API
4. SQL
5. Rest API
6. Github
7. Doxygen

Full Description

Movesense is an open development environment for motion sensing solutions. Its easy and innovative tools allow you to build your own wearables quickly and cost-efficiently.

Movesense sensor is the most versatile sports sensor in the world: You can develop your own app in the sensor to make it work exactly as you need. Its capable sensing, robust construction, and tiny size make it perfect for measuring any sports and much more.

With Movesense, you can track sports, health, equipment, machinery – basically anything that moves. Created by the Finnish sports watch expert Suunto, it is your fast access to proven sensor hardware and technology.

Exceptional for a small battery powered device, the sensor runs Movesense Device API, a REST-type service framework for building applications in the sensor and to provide data to phone apps.

You must use the Movesense to build a wearable that can be worn by the elderly to track their health and wellbeing.

Mandatory

Application Area Identified Identification of area for applying wearable;

Data Reading Read and Store raw data from the sensor(s);

Display Relevant data displayed visually for users;

Discretionary

Management System contains management software to allow for multiple users.

Extraction Relevant and useful information is extracted from the raw data and shown in a user friendly manner.

Exceptional

Portable App runs across multiple platforms (Android, PC, Mac);

Releasable App is of releasable quality.

Chapter 4

The Development Process

The project is completed using an iterative process. This is done to replicate, as closely as possible, software development in the real world. There are four iterations and marks are awarded at the end of each iteration. The iterations are divided as follows:

Inception Phase

The first iteration is the inception phase. The student does research on their project and starts on their function specification. At the end of this iteration the student will understand more fully what the project entails and the technologies involved.

Deliverables

Project Plan Full plan outlining what will be delivered (in what order) in the following iterations.

Poster An A2 sized poster summarizing the project.

Initial Function Specification The initial specification will contain:

- Project title and description;
- Context Diagram;
- Project Requirements;
- Users Identified.

Project Github Setup The project git (or equivalent code versioning system) will be set up fully and access will be granted to the project

supervisor. Git will contain a readme file and the licence information at a minimum.

Alpha Phase

Deliverables

Completed Functional Specification The following sections of the Functional Specification will be completed:

- Fully defined Use cases;
- Full FURPS analysis;
- Metrics will be completed.

Presentation Students will give a five minute presentation to the class and supervisors outlining their progress.

Initial Design Document This design document release will contain:

- Architectural Definitions - description of the technologies used in the project (programming language, database, frameworks);
- Project Plan;
- UI/UX of the product.

Alpha Release The first code release will be available on the git repository. It will prototype important functionality as agreed by the student and their supervisor. This release will show the feasibility of the end product.

Release Candidate Phase

Deliverables

Beta Release A standalone executable must be demonstrated that shows a working release (even if there may be a lot of missing functionality at this stage). The scope of this release will be negotiated with the supervisor.

Completed Design Document The completed Design document will contain:

- Pseudocode for any important and/or complex algorithms required;
- Full data definitions (SQL, noSQL);
- Class Diagram (UML)
- Message Sequence Diagrams.

Final Release

Deliverables

Project Report The final project report will be submitted. This must be conform the the report definition as laid out in the document below.

Release A final working release of the product. A link to the git repository containing the code must be available to supervisors. The codebase must also be submitted seperately in a zip file.

Chapter 5

The Examination Process

Final Demonstration

Some time after the final submission date of the project materials you will demonstrate your project to the panel of tutors. The demonstration will normally commence with a short verbal outline of the goals and purpose of your project, as most of the panel will not be familiar with it. Part of the demonstration will normally include installation of your software on the target platform from the usb key (or other media) you have submitted. By prior arrangement with your mentor you may be permitted to work with pre-installed materials. Following your demonstration, during which you will not be questioned (except for requests for demonstration of particular features, clarifications etc) you will be questioned on your different aspects project by the panel.

This procedure should take approximately twenty minutes (ten minutes for the demonstration and ten for questions). It is up to you to make sure that you demo your FYP in a manner that displays it in the most favourable light. Always rehearse your demo in advance and make sure that everything will install and work on the *actual* demo equipment.

Some tips:

- Always do a rehearsal of the demo in the actual room where the demo will take place using the equipment in that room;
- Always make sure (in advance of your demo) that your software will install and work **in** the demo room and **on** the demo equipment;
- Always have your demo timed to last ten minutes otherwise you will not get to demonstrate every feature of your project and may lose marks;

- Always rehearse in front of someone who will spot any issues (bad explanations etc.) and who will ask questions that the supervisors might ask (use your classmates for this);
- Arrive at the demo early;
- It is normal to be nervous. Remember we do not ask questions to catch you out but to try make sure we understand your project and how much work you have put into it.

Project Mark Allocation

Project elements weighting is as follows:

Functional Specification 5%

Research Poster 5%

Design Manual 5%

Presentation 5%

Project Report 10%

Usability 10%

Project Quality 20%

Final Product 30%

Process 10%

Always remember how many marks are going for each report especially when deciding how much work to put into it.

Schedule of Marks

Marks will be allocated throughout the year. The project is divided into four iterations over the two semesters. Marks will be allocated as follows:

Middle Semester One 10%

End Semester One 15%

Middle Semester Two 20%

End Semester Two 55%

Table 5.1: Schedule of Marks

Iteration	Functional	Design	Poster/Talk	Report	Process	Usability	Quality	Product	Total
Inception	2.5	0	5	0	2.5	0	0	0	10
One	2.5	2.5	5	0	2.5	0	0	2.5	15
Two	0	2.5	0	0	2.5	0	5	10	20
Three	0	0	0	10	2.5	10	15	17.5	55
Total	5	5	10	10	10	10	20	30	100

Rubrics

The rubrics listed here are provided to serve as guidelines to help you assess the standard of your work. Each project is unique and is marked on its own merits however the rubrics show what we will expect from all the deliverables.

Table 5.2: Research Poster

	Fail (0-19)	Pass (40-59)	Good (50-69)	Excellent (70-100)
Standards	Formatting is inconsistent. Standards ignored.	Inconsistent following of standard and/or inconsistent formatting.	Format is good but does not follow standard. Print quality is bad.	Documents follows all standards completely and consistently
Writing	Basic Spelling and grammar errors. No structure to document.	Structure is poor. Spelling and grammar fair. Difficult to follow reasoning of document. Sentences unclear.	Document narrative is difficult to follow. Spelling and grammar good overall. Lack of focus in text.	Document is well written. No spelling or grammar mistakes. Poster is well structured and easy to follow.
References	No references and/or no citations	References/Citations are not relevant or add little to poster	References are relevant but not complete.	References are relevant, complete and cited. References are mainly published and/or peer reviewed works.
Depth/Breath	Does not cover appropriate topics.	Most topics of relevance are covered. Topics are covered in some depth but not enough to draw any relevant conclusions.	All topics covered but lacking either in depth or in breath.	All aspects of the project are covered. Poster is of conference quality
Summary	No conclusion	Summary not related to research	Summary is vague and/or unclear. Conclusions do not follow from poster text	All research is brought together and summarized. Good conclusions.

Table 5.3: Functional Specification

	Fail (0-39)	Pass (40-59)	Good (60-70)	Excellent (70-100)
Standards	No page numbers. No table of contents. Formatting is inconsistent. Standard ignored	Inconsistent following of standard and/or inconsistent formatting.	Format is good but does not follow standard. Print quality is bad.	Documents follows all standards completely and consistently
Writing	Basic Spelling and grammar errors. No structure to document	Structure is poor. Spelling and grammar fair. Difficult to follow reasoning of document. Sentences unclear.	Document narrative is difficult to follow. Spelling and grammar good overall. Lack of focus in text .	Document is well written. No spelling or grammar mistakes. Sentences are clear. Document is well structured and easy to follow.
Specification	Context Diagram missing. Users not identified. Core functionality not identified	Some Use Cases missing. Not all core functions or users identified.	Users identified. Context Diagram complete. Use cases all present but lacking in clarity/detail	Context Diagram identifies all core functions. All users identified Use Cases for all interactions identified in the Context Diagram.
Metrics	No FURPS or Metrics	Metrics not precise enough to be useful or are irrelevant.	Metrics are useful but not complete. FURPS lacking detail.	Full FURPS included. Valid and complete metrics Identified
Requirements (scope, aims and definition)	Requirements (or Research Question) not identified	Some requirements identified but not with sufficient detail for design phase. No justification provided.	Most requirements identified. Some detail lacking.	Precise Requirements identified. All libraries and tools needed for project identified. All requirements are justified in document

Table 5.4: Design Document

	Fail (0-39)	Pass (40-59)	Good (50-69)	Excellent (70-100)
Standards	No page numbers. No table of contents. Formatting is inconsistent. Standard ignored.	Inconsistent following of standard and/or inconsistent formatting.	Format is good but does not follow standard. Print quality is bad.	Documents follows all standards completely and consistently
Writing	Basic Spelling and grammar errors. No structure to document	Structure is poor. Spelling and grammar fair. Difficult to follow reasoning of document. Sentences unclear	Document narrative is difficult to follow. Spelling and grammar good overall. Lack of focus in text	Document is well written and structured. No spelling or grammar mistakes. Sentences are clear.
Design	Design is missing or unusable.	Class Diagram is simplistic and/or does not match project specifications. Use cases lacking detail. No Message Sequence Diagrams or critical algorithms identified	Incomplete or difficult to read class diagram. Some use-cases missing detail. Some incomplete Message Sequence Diagrams. Incomplete architecture design	Detailed Use Cases clearly presented. Full UML Class Diagram. Message Sequence Diagrams for all functions. Pseudocode for critical algorithms. Clear Architecture model.
Methodology	No methodology present	Methodology not precise and not verifiable	Methodology valid but unclear in parts	Valid and verifiable methodology presented. Methodology is complete and customised to project aims. Theoretical analysis presented
Plan	No plan.	Unrealistic plan. Milestones missing.	Plan contains most milestones. All phases identified but not fully or realistically scheduled.	Detailed and Realistic Plan presented. All Milestones, Deliverables and Development phases identified and scheduled.
UIX	No defined UI.	UIX lacks enough detail to allow for implementation. Screenshots shown but not clear how screens work together.	Incomplete UIX documented. No or unconvincing rationale for UIX design.	Full UIX documented. Rationale for UIX shown.

Table 5.5: Project Report

	Fail (0-39)	Pass (40-59)	Second Class Honours (50-69)	First Class Honours (70-100)
Standards	No page numbers. No table of contents. Formatting is inconsistent. Standard ignored.	Inconsistent following of standard and/or inconsistent formatting.	Format is good but does not follow standard. Print quality is bad.	Documents follow all standards completely and consistently.
Writing	Basic Spelling and grammar errors. No structure to document	Structure is poor. Spelling and grammar fair. Difficult to follow reasoning of document. Sentences unclear	Document narrative is difficult to follow. Spelling and grammar good overall. Lack of focus in text	Document is well written. No spelling or grammar mistakes. Sentences are clear. Document is well structured and easy to follow.
Reflection	No indication of reflection by the student.	Report shows some reflection but only at a shallow level.	Report shows reflection and insight by the student but is not presented in a clear and accessible manner.	Clearly identifies problems encountered during project. Clear and in-depth sections on what was achieved, what was not achieved, what was learned and what would be done differently if the project started over.
Final Design	No final design shown.	No testing based on the metrics from the design manual.	Final Design not clearly defined. Tests are incomplete.	Final Data structure design shown clearly (ERD, etc). Results of testing the project using the Design manual metrics listed clearly. Performance metrics and simulation results presented clearly
Conclusions	None provided.	Vague or incorrect conclusions.	Conclusions not fully substantiated.	Valid and well argued conclusions presented.

Table 5.6: Usability

	Fail (0-39)	Pass (40-59)	Second Class Honours (60-70)	First Class Honours (70-100)
UIX	No evidence of UIX design	UIX is badly designed or incomplete	UIX is designed and good but incomplete (beta quality).	Highly usable product. UI is complete
Documentation	None.	Incomplete and bad documentation.	Good documentation provided but incomplete.	Full documentation provided for user training. Documentation is almost release quality (Screen-cast, Video, Manual,...)
Metrics	No user testing.	Some informal user testing.	User testing performed but feedback not properly collected or not incorporated.	User testing has been completed and feedback incorporated in UIX.

Table 5.7: Process

	Fail (0-39)	Pass (40-49)	Second Class Honours (60-70)	First Class Honours (70-100)
Diary	No Dairy	Diary is badly written or incomplete.	Diary is good but incomplete. Some missing weeks	Diary is complete and well written. Each week signed.
Process	No process. No Code Version System used.	Process intermittently and improperly used. Code changes not tracked.	Appropriate Process employed (Agile, etc.). Implementation on CVS (Git, Svn, ...) but not properly used.	Process employed correctly. Implementation all on CVS (Git, Svn, ...) with evidence of proper usage (branches, releases, etc.)
Engagement	No engagement with Supervisor.	Some informal engagement.	Feedback from supervisor not properly collected or not incorporated.	Feedback properly incorporated in process.

Table 5.8: Presentation

	Fail (0-39)	Pass (40-59)	Good (60-70)	Excellent (70-100)
Slides	Slides not readable or not on topic	No screencast. Slides not well formatted. Incomplete or irrelevant information on slides.	Poor screencast – not relevant or not well done	Slides are consistently formatted. Slides are clear and readable. Slides include an edited screencast showing project in action.
Speaking	Difficult to understand. No interaction with audience. Too long or too short.	Talk goes off topic. Bad interaction. Difficult to understand speaker.	Talk slightly too long or too short. Questions not clearly addressed.	Good timing (5 minutes). Clear articulation. Good Audience interaction. Questions all addressed well. Student has incorporated feedback from first talk into second talk to make visible improvements
Breath/Depth	Off topic.	Talk not always on topic. Does not cover all relevant details.	Talk is on topic. Missing depth in key areas	Covers project in sufficient detail. Identifies core issues/problems with project

Table 5.9: Product Quality

	Fail (0-39)	Pass (40-59)	Good (60-70)	Excellent (70-100)
Code Standard	Does not follow any standard.	Code follows a standard but is inconsistent. Difficult to read or follow code.	Code follow appropriate standard. Overall structure could be improved.	All code follows appropriate coding standard. Code easy to follow and read. Good overall structure.
Correctness	Code does not work.	No tests provided to check code correctness.	Code appears to work but missing or incomplete test cases.	Code works as specified. Handles all edge cases. Tests included in code base.
Documentation	No Documents.	Documented using inline and header comments. No separate documentation provided	Documents provided by tool (e.g. Doxygen) but tool not used properly. Documentation incomplete or not well structured	All code documented properly using tool such as Doxygen. Full code documentation provided in various formats (HTML, pdf, etc.). Documentation well structured, clear and precise

Table 5.10: Amount Achieved

	Fail (0-39)	Pass (40-59)	Good (60-70)	Excellent (70-100)
Difficulty	The project is a simple CRUD application. The project was developed using technologies already well known to the student.	Well established development platforms (libraries, APIs etc.) were required. The project contains features beyond simple CRUD applications	The new technologies required more than just applying API calls to work. Two or more APIs were used.	New leading edge technologies were employed successfully. Two or more different API's/languages were combined seamlessly. The software system runs across multiple platforms. New or very complex algorithms were written from scratch. The system was developed using languages, environments, etc. that were unknown to the student before the project.
Completeness	The project is not working	The project is not at a beta release stage. The project works only from inside the development IDE.	The project is a working stand alone beta release. Full install files are included	The project achieved all of the goals set out in the project specification. All components work together properly. A polished working release is completed.
Testing and Metrics	No metrics or tests were applied.	Evidence of some testing but the tests are not compliant with the metrics.	Some testing has been applied based on the metrics	The project has completed a full suite of tests based on the project metrics.
Implementation	Little or no original implementation.	Some features are implemented but mainly uses other peoples libraries. Original implementation is simplistic.	Most/all implementation is original.	Most/all implementation is original. Implementation is complex.

Chapter 6

Deliverables

Research Poster

This is the one of the first deliverables of the project. Upon completion of this report you will have discovered enough information to make most of the important decisions for the rest of the FYP. You will have reached conclusions on your implementation tools (languages, databases, etc.), platforms (Operating Systems, hardware components, etc.) and any required algorithmic approaches.

Once this is complete you should be confident about what you need to do. Moreover, and perhaps more importantly, the poster should demonstrate to anyone who reads it why you have made those decisions. This report needs to demonstrate that you have achieved mastery of the topics involved in your project.

It is not enough for you to know the answers, the document must also convince anyone who reads it that you know the answers. If it does not do that then it has failed. The poster will be on display on the end of year industry demo day and will be viewed by potential employers.

The research poster should have some or all of the following sections:

Abstract One paragraph in length. It is a summary of the entire project;

Introduction An overview of everything that is to follow in the poster;

Overview of areas, technologies or topics researched For each area, technology or topic:

- Reasons for researching and perceived relevance to project;
- What you discovered during research;

- Utility of the research and any further research it suggested;

Summary and Conclusions A review of everything that preceeded this section. It finishes with concret conclusions gleaned from the report;

Appendix Detailed algorithm descriptions that you plan to implement;

Glossary Full list of all acronyms used in the report;

Bibliography List of all references cited in the report. If you get this wrong it can invalidate the entire report. There are tools that will create this for you - **use them!**.

Remember the poster will by definition be a short summery of your project. You will have to decide, alongside your supervisor, what to put in and what to leave out. There are many examples of research posters on the internet. your poster must be of conference quality.

Functional Specification

Define the Application.

- What is the application supposed to be?
 - What the product is.
- What will the application do?
 - Usually a small number of core pieces of functionality and a larger number of less important functions;
 - List all functionality- re arrange with most important first;
 - It is considered good practice to identify each feature in terms of whether it is a core feature (must have) or a non-core feature (nice to have but not essential). You need to complete the core features to pass.
- Who is going to be using this application?
 - Identify the user group(s).
- Context Diagram and Use Cases
- Metrics

- How will you (and we) gauge if your project is successful.
- Is there a precedent for this application? (your inspiration):
 - Is it similar to something already in existence?
 - How does it differ?

Project Plan

- Introduction:
 - What are you planning for?
 - Requirements: Software and Hardware.
- Major milestones in chronological order:
 - To include but not be limited to deliverable dates;
 - Before you fill this in, roughly plan - with final presentation date first, work back to now; think how long before this that you will have completed your final installation and testing. How long before this to you next furthest milestone etc.;
 - Any non-intuitive topics should be explained in tabular form, all topics should be included in Gantt Chart or using Microsoft Project;
- Any Exceptional requirements, for example:
 - e.g. meet with client/user;
 - visit client location;
 - depends upon library otherwise will have to implement our own

Design Manual

- Model the Application.
- Design and Describe how the application will be used:
 - e.g. Navigation sequence in Web based project;
 - This may necessitate some high level design in order to convey how the application may be used to access its functionality;

- User interfaces should be specified loosely here (possibly screen shots of prototype user interfaces - you will not be held to using these UIs they are just to assist in explanation of your project functionality);
- Major architectural components and their relationship to each other should also be diagrammatically (with brief explanation paragraphs) presented here e.g Database backend, web server front end with ODBC connecting them etc.;
- Include any element that you would feel conveys how you perceive your project will function.

Web Pages

Each project needs to have an associated website. Hosting is provided at <https://showcase.itcarlow.ie/>. The website will contain all the project documentation along with information about the project. This information usually consists of text alongside a demo of the project (as a screencast). The site is advertised to potential employers and is used as part of the end of year showcase when employers from all over Ireland come down to view the FYP's and talk to you. These web sites are important as it is not uncommon for students to be offered jobs at the showcase event based on their FYP. Last years showcase websites are available online for you to see what is required.

Project Presentation

You should prepare your presentation in powerpoint (or similar e.g. $\text{\LaTeX}$). Each presentation is five minutes in length followed by five minutes for questions. All the supervisors and the rest of the class will be present at the presentations. You should dress appropriately (smart casual or better) for this event. During the presentation you will explain to your peers and all the supervisors what, in essence, your project is about and how it is proceeding.

The presentation is given twice. The first time you will be marked and given feedback on how your talk went. This allows you to learn from the first presentation. The second presentation will be similar to your first presentation but you will use the feedback from the first presentation to improve your talk. It will occur at the end of the third iteration. The marks from your second presentation will be applied to your FYP mark.

Given the feedback from the first presentation we expect that the second presentation will be much improved and the overall marks for second pre-

sentation should high. The presentation feedback will also include tips from other mentors on how to improve your project and/or any potential issues that they may have spotted.

User Manual

Not all applications require a user manual, e.g. phone apps do not generally require one. In fact, in some cases, applications should **not** require a user manual. Check with your mentor to see if a manual is required.

The user manual can take the form of an online video demonstrating usage or an online website tutorial. The language used in the manual should be chosen to match the target user. The level of explanation and phraseology in a game project, for example, will not be the same as that chosen for the manual for a network testing tool. The user manual should generally not present tutorial material on the project topic, but references to such material may usefully be included. The manual should include:

- System requirements;
- Installation instructions;
- System usage, with screenshots where needed.

Project Report

The project report is the last report delivered during FYP. This report should demonstrate that you have learnt new skills during the development of your project. if you made mistakes during the development here is where you show that you have identified those errors and are aware of them. In fact we expect you to make mistakes during the development process but good students learn from their mistakes while bad students do not. The project report should show us that you are a good student, that is, that you have learnt something while working on the FYP and that you are now a more capable software developer as a result.

Below are the questions that the report should answer. Each question can be given its own section heading. When answering be as specific as possible - remember you need to demonstrate that you now know how to undertake a large development process.

General Issues

- Problems encountered and how they were resolved. Be as specific as possible here;
- What you achieved;
- What you did not achieve;
- What you learned;
- What you would do differently if starting again;

Technical Issues

- Report of any differences from your earlier design, and any additional research that was required;
- Module descriptions;
- Data structures including any files and/or databases and their record layouts;
- Testing used to assess reliability of your software product.

Code Listings

Code should comply with the Institute Standards. It is advisable to use a tool such as *Doxygen* (or e.g. *JavaDoc*) to automatically create formatted code listings for you. As a general rule manually created code listing not only score much lower than automatically generated ones but are also much harder to create. A tool such as Doxygen will create a full professional level document with just a few mouse clicks.

Project Software

Your project code should be delivered on memory stick as appropriate. It should be auto-installing. There should be a separate directory on the disk called “Docs” which includes copies of all documentation submitted during the year, presentation (powerpoint), source code etc. If non-standard additional software (e.g. components for compilers, tool kits etc) are required to build your project, copies should be included in an appropriate directory.

All project code must be hosted on Github or equivalent CVS (see your supervisor for acceptable alternatives).

You must make this repository available to your supervisor from the start of the project.

Chapter 7

Document Standards

Introduction

Supplements are reference items generally added to a long report or proposal to make the document more accessible. A document's supplements accommodate readers with various interests: the title page, table of contents, and abstract give summary information about the content of the document. The glossary, appendixes, and reference list either provide supporting data or help readers follow technical sections. All supplements are written only after the document itself has been completed.

Each page (other than the title page) of each document should have a header with the project title and document name and a footer with page number and page count (e.g. 3/15). Any document more than 3 pages long **must** have a table of contents and should be presented suitably bound, with a transparent sheet over the front cover.

Document text should normally be font size 10 or 12.

Cover

Use a sturdy cover (plain). Centre the report title and your name four to five inches from the upper edge. Include the type of document that follows (e.g. Technical Manual or Research Manual, etc.):

Automatic Licence Plate Recognition
Research Manual
by
XXX

Title Page

The title page lists:

- The report title;
- Author's name;
- Name of person(s) or organisation to whom the report is address;
- Date of submission.

Title. Your title announces the report's purpose and subject. It should be clear, accurate, comprehensive, and specific.

Placement of title page items. Do not number your title page, but count it as page i of your prefatory pages. Centre the title horizontally, three to four inches below the upper edge, using all capital letters.

Abstract

Readers may not have the time to read all of your report. For these an abstract is the most important part of the document. The abstract is always written after your report.

Follow these guidelines for your abstract:

- Make your abstract stand alone in meaning.
- Write for general readers. Translate technical data.
- Add no new information. Summarise the report.
- Present your information in the sequence:
 - Identify the issue or need that led to the report.
 - Offer the major findings from the report body.
 - Include a condensed conclusion and recommendations, if any.

Acknowledgements

Acknowledge those who helped you with the report (separate page). use them!

Table of Contents

Your table of contents serves as a road map for readers and a checklist for you. Compose this supplement by assigning page numbers to headings from your outline.

Follow the following guidelines:

- List front matter (acknowledgements, abstract) numbering the pages with small *roman numerals*. The title page is not listed. List glossary, appendix, and endnotes; number these pages with *Arabic numerals*, continuing the page sequence of your report proper. The first page of report text is page 1.
- Don't include headings in the table of contents not listed as headings or subheadings in the report. The report may contain subheadings not listed in the table of contents.
- Phrase headings in the table of contents exactly as in the report.
- To reflect their rank, list headings at various levels in varying typeface and indentation.
- Use dots (.....) to connect heading to page number. Align page numbers vertically above each other.

Table of Tables and Figures

Following the table of contents is a table of tables and figures, if needed. When a report has more than four or five visuals, place this table on a separate page.

Glossary

A glossary is an alphabetical listing of specialised terms and their definitions, following your report. A glossary makes key definitions available to nontechnical readers without interrupting technical readers.

Guidelines for a glossary:

- Define all terms unfamiliar to a general reader.
- Define all terms that have a special meaning in your report.
- List your glossary and its first page number in your table of contents.

- List all terms in alphabetical order. Underline each term and use a colon to separate it from its single spaced definition.
- Define only terms that need explanation.
- On first use, place an asterisk in the text by each term defined in the glossary.

Appendixes

An appendix follows the text of your report. The appendix expands items discussed in the report without cluttering the report text.

Typical items to include in an appendix:use them!

- Details of an experiment;
- Statistical or other measurements;
- Maps;
- Complex Formulas;
- Long quotations (one or more pages);
- Photographs;
- Related correspondence (letters of inquiry);
- Interview questions and responses;
- Sample questionnaires and tabulated responses;
- Sample tests and tabulated results;
- Visuals occupying more than a page;
- Anything more essential to secondary readers than to primary readers.

The appendix is a catch all for items that are important but difficult to integrate within your text. Do not stuff appendixes with needless information.

More guidelines:

- Include only material that is relevant.
- Use a separate appendix for each major item.

- Title each appendix clearly: “Appendix A: Code Listing”
- Do not use appendixes excessively.
- Refer readers to it at appropriate points in the report: “(see Appendix A)”.

Use the appendix for any material that is essential, but might harm the coherence of your report. Readers should be able to understand the report without having to turn to the appendix.

Ex The whale population declined drastically between 1986 and 1987 (see Appendix B for details).
{Improper reference}

The whale population declined by 16 percent from 1986 to 1987 (see Appendix B for statistical breakdown).
{Proper reference}

Bibliography

You should acknowledge your debt to each information source you used. Proper documentation satisfies professional requirements for ethics, efficiency, and authority (i.e. in making a claim).

What you should document

You must document:

- Any source from which you use exact wording;
- Any source from which you adapt material in your own words;
- Any visual illustration: charts, graphs, drawings etc.

How you should document

The bibliography should be in alphabetic order. Most scientific publications use the name-and-year system in which authors names and the date of publication are cited in the text ?. The college uses the Harvard referencing standard so this must be used in all documents.

Different sources are cited differently so articles such as ??? and ? will have a different format in the bibliography than, for example, technical reports such as ??, proceedings like ? and books e.g. ???.

There is even a format used for miscellaneous articles that do not fit into the above categories, see ?? and web references such as ?.

Full Referencing Guideline Document

There is a separate document telling you how to reference correctly. It is called **Credit where Credit is Due** and is on this blackboard site. Please use this for referencing guidelines.

Chapter 8

Conclusion

The FYP is the most important part of your fourth year. It is here that you demonstrate that you are a capable Software Developer. It takes up one third ($\frac{1}{3}$) of the final year of the course and the marks you get here will have the greatest influence on your final awarded grade. Employers will use the FYP to decide whether to give you a job or not. The hours you need to spend on the FYP are not explicitly scheduled on your timetable but you need to spend one third of your time during the year on your FYP!

The set of deliverables and the deadlines for their delivery will be given to you at the start of the year. You must keep to those deadlines. Remember that your workload will increase throughout the year not decrease. Try to do as much work on your project as early as possible to compensate for this.

Once you are assigned a FYP you are expected to take ownership of that project. Your assigned supervisor is there to guide you through the development process but you must do the development. The project is your project not your supervisors.

There is a parable used to help developers understand the Agile process.

A chicken and a pig decided to open a restaurant selling egg and bacon breakfasts. When they came to name the restaurant the chicken wanted to name it *Chicken and Pig's Hearty Breakfasts* but the pig refused saying instead that it should be called *Pig and Chicken's Hearty Breakfasts*. When the chicken asked why the pig's name came first the pig said: *It's like this, you are only involved in this project but I am committed!*

You would do well to remember this. While supervisors are involved in your FYP it's your bacon that is on the line.